

MAT 362 HW07

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Problem. 3.2.1. Show that at a hyperbolic point, the principal directions bisect the asymptotic directions.

Solution. The locus of points on the Dupin indicatrix are the points $ae_1 + be_2$ such that $k_1a^2 + k_2b^2 = 1$ or $k_1a^2 + k_2b^2 = -1$ where e_1, e_2 are the principal directions and k_1, k_2 the principal curvatures; this is a union of two hyperbolas whose (shared) asymptotes are the asymptotic directions at p .

Since the axes of symmetry of the hyperbolas bisect their asymptotes, we just need to check that the axes of symmetry of $k_1a^2 + k_2b^2 = 1$ are the principal directions (shared asymptotes means that the axes of symmetry are the same for the other hyperbola). This is obvious, though: e_1 and e_2 are the axes of symmetry if and only if the maps $a \mapsto -a$ and $b \mapsto -b$ both preserve the hyperbola, and this is trivial because $a^2 = (-a)^2$ and $b^2 = (-b)^2$. $\square$

Problem. 3.2.3. Let $C \subseteq S$ be a regular curve on a surface S with Gaussian curvature $K > 0$. Show that the curvature k of C at p satisfies

$$|k| \geq \min(|k_1|, |k_2|),$$

where k_1 and k_2 are the principal curvatures of S at p .

Solution. Let k_n be the normal curvature of C at p . Since $K = k_1k_2$ is positive, k_1 and k_2 have the same sign. The principal curvatures are the extremal normal curvatures among all normal curvatures at p of curves in S , so k_n lies between k_1 and k_2 , hence k_1 and k_2 having the same sign means that $|k_n| \geq \min(|k_1|, |k_2|)$. In turn, we have

$$|k| \geq |k \cos \theta| = |k_n| \geq \min(|k_1|, |k_2|)$$

where $\theta = \langle n, N \rangle$, where n and N are the (unit) normal vectors to C and S respectively at p . $\square$

Problem. 3.3.1. Show that at the origin $(0,0,0)$ of the hyperboloid $z = axy$ we have $K = -a^2$ and $H = 0$.

Solution. Let $p = (0,0,0)$ and observe that the hyperboloid S is parametrized by $\mathbf{x}(u,v) = (u,v,auv)$. We have $T_p(S)$ has basis $\mathbf{x}_u = (1,0,av) = (1,0,0)$ and $\mathbf{x}_v = (0,1,au) = (0,1,0)$. For a unit vector $(s,t,0)$ in $T_p(S)$, the plane through $(s,t,0)$ and $N(p) = (0,0,1)$ intersects S in the points (sx,tx,y) satisfying $y = astx^2$, i.e. the curve $\alpha(x) = (sx,tx,astx^2)$. The curvature of this curve at p is

$$\frac{|\alpha' \wedge \alpha''|}{|\alpha'|^3} = \frac{|(s,t,0) \wedge (0,0,2ast)|}{|(s,t,0)|^3} = |(2ast^2, -2as^2t, 0)| = |2ast|.$$

The normal to this curve at p is

$$\left[\frac{d}{dx} \frac{\alpha'(x)}{|\alpha'(x)|} \right]_p = \frac{1}{|2ast|} (0,0,2ast),$$

which is either $(0,0,1)$ or $(0,0,-1)$ depending on if $2ast$ is positive or negative respectively, while the normal to S at p is just $(0,0,1)$. This means the normal curvature at p is $2ast$ (either $|2ast| \cdot 1$ if $2ast$ is positive or $|2ast| \cdot -1$ if $2ast$ is negative). Since we have pre-emptively fixed $s^2 + t^2 = 1$ this means $2st$ is within the interval $[-1,1]$ because $1 = s^2 + t^2 \geq |2st|$ with equality iff $|s| = |t|$ by the AM-GM inequality, so the principal curvatures are $\pm a$. Therefore, $K = -a \cdot a = -a^2$ and $H = \frac{-a+a}{2} = 0$ as desired. $\square$

Problem. 3.3.5. Consider the parametrized surface

$$\mathbf{x}(u, v) = \left(u - \frac{u^3}{3} + uv^2, v - \frac{v^3}{3} + vu^2, u^2 - v^2 \right),$$

known as **Enneper's surface**, and show that

(a) The coefficients of the first fundamental form are

$$E = G = (1 + u^2 + v^2)^2, \quad F = 0.$$

(b) The coefficients of the second fundamental form are

$$e = 2, \quad g = -2, \quad f = 0.$$

(c) The principal curvatures are

$$k_1 = \frac{2}{(1 + u^2 + v^2)^2}, \quad k_2 = -\frac{2}{(1 + u^2 + v^2)^2}.$$

(d) The lines of curvature are the coordinate curves.

(e) The asymptotic curves are $u + v = \text{const.}$ and $u - v = \text{const.}$

Solution. (a) We have $\mathbf{x}_u = (1 - u^2 + v^2, 2uv, 2u)$ and $\mathbf{x}_v = (2uv, 1 - v^2 + u^2, -2v)$, so

$$\begin{aligned} E &= (1 - u^2 + v^2)^2 + (2uv)^2 + (2u)^2 \\ &= 1 + u^4 + v^4 - 2u^2v^2 - 2u^2 + 2v^2 + 4u^2v^2 + 4u^2 \\ &= 1 + u^4 + v^4 + 2u^2v^2 + 2u^2 + 2v^2 \\ &= (1 + u^2 + v^2)^2. \end{aligned}$$

The calculation for G is symmetric (swap u and v). Finally,

$$\begin{aligned} F &= 2uv(1 - u^2 + v^2 + 1 + u^2 - v^2) - 4uv \\ &= 2uv \cdot 2 - 4uv = 0. \end{aligned}$$

(b) We first calculate

$$\mathbf{x}_u \wedge \mathbf{x}_v = (-2u(u^2 + v^2 + 1), 2v(u^2 + v^2 + 1), 1 - (u^2 + v^2)^2) = (u^2 + v^2 + 1) \cdot (-2u, 2v, 1 - u^2 - v^2)$$

and

$$\begin{aligned} |\mathbf{x}_u \wedge \mathbf{x}_v| &= (u^2 + v^2 + 1) \sqrt{4u^2 + 4v^2 + (u^2 + v^2 - 1)^2} \\ &= (u^2 + v^2 + 1) \sqrt{u^4 + v^4 + 1 + 2u^2 + 2v^2 + 2u^2v^2} \\ &= (u^2 + v^2 + 1)^2, \end{aligned}$$

so the normal is

$$N = \frac{1}{u^2 + v^2 + 1} (-2u, 2v, 1 - u^2 - v^2).$$

Now we can compute

$$\begin{aligned} e &= \langle N, \mathbf{x}_{uu} \rangle = \frac{1}{u^2 + v^2 + 1} ((-2u)^2 + (2v)^2 + 2(1 - u^2 - v^2)) = 2 \\ g &= \langle N, \mathbf{x}_{vv} \rangle = \frac{1}{u^2 + v^2 + 1} ((-2u) \cdot 2u + (-2v) \cdot 2v - 2(1 - u^2 - v^2)) = -2 \\ f &= \langle N, \mathbf{x}_{uv} \rangle = \frac{1}{u^2 + v^2 + 1} ((-2u) \cdot 2v + 2v \cdot 2u) = 0. \end{aligned}$$

(c) The principal curvatures are $H \pm \sqrt{H^2 - K}$, which can be expanded as

$$\frac{1}{2} \frac{eG - 2fF + gE}{EG - F^2} \pm \sqrt{\left(\frac{1}{2} \frac{eG - 2fF + gE}{EG - F^2}\right)^2 - \frac{eg - f^2}{EG - F^2}}.$$

We have $2fF = 0$ and $e = -g$, $E = G$ implying $eG + gE = 0$, so $eG - 2fF + gE = 0$. Thus the principal curvatures are

$$\pm \sqrt{-\frac{eg - f^2}{EG - F^2}} = \pm \sqrt{-\frac{eg}{EG}} = \pm \sqrt{-\frac{2 \cdot (-2)}{(1 + u^2 + v^2)4}} = \frac{\pm 2}{(1 + u^2 + v^2)^2}.$$

(d) Every point of the surface is not umbilical because part (c) shows that the principal curvatures are always different, so, since $F = f = 0$, the coordinate curves are the lines of curvature, as stated on page 163.

(e) The asymptotic curves are the curves $\alpha(t) = \mathbf{x}(u(t), v(t))$ satisfying the differential equation

$$e(u')^2 + f u' v' + g(v')^2 = 2(u')^2 - 2(v')^2 = 2(u' + v')(u' - v') = 0$$

for all $t \in I$ (where α is a map $I \rightarrow \mathbb{R}^3$). We deduce that either $u' = v'$ always holds or $u' = -v'$ always holds. The only alternative is that at some point either u' or v' changes sign (say WLOG u') while the other doesn't change sign; this implies that $u'(x) = 0$ for some $x \in I$, which then also implies $v'(x) = 0$, but then $\alpha'(x) = (0, 0, 0)$, which contradicts the regularity of α . Thus $\alpha(x) = (u(x), c + v(x))$ for all $x \in I$ or $(u(x), c - v(x))$ for all $x \in I$ where c is some constant; either way, α can be reparametrized as one of the lines $u + v = c$ or $u - v = c$ as desired. $\square$